FINAL EXPLANATION: CORE MATHEMATICS GRADE 10

Student Assessment Document

FINAL EXPLANATION: CORE MATHEMATICS – GRADE 10 | SUB-STRAND 2.8: VECTORS	
Student Assessment Document	
Student Name	____
Class	____
Date	____

INSTRUCTIONS FOR STUDENTS

This is your Final Explanation task. You have spent 8 lessons investigating why two matatu journeys of exactly the same distance from Nairobi's CBD end up at completely different destinations. Now it is time to bring everything together and answer the driving question fully and precisely.

READ THE DRIVING QUESTION CAREFULLY:

"Why do two matatu journeys of exactly the same distance from Nairobi's CBD end up at completely different destinations — and what mathematical tool do we need to describe a journey completely?"

YOUR TASK:

Write a complete, well-organised explanation that answers the driving question using the mathematical ideas you have learned in this unit. Your explanation must move logically from the phenomenon (the two matatus) all the way to the mathematical tool (vectors) and show how that tool solves the puzzle.

Work through each section below in order. Use the sentence starters if you need help getting started, but you are encouraged to go beyond them. Support every claim with a mathematical example, diagram description, or calculation.

TIPS FOR A STRONG RESPONSE:

- Always connect your mathematics back to the matatu scenario or another real Kenyan context.
- Use correct mathematical notation (e.g., vector AB written as AB with an arrow, column vectors, magnitude notation |v|).
- Show working where calculations are required.
- Diagrams or sketches are encouraged — describe or draw them clearly.
- Aim for precision: a good scientist or mathematician says exactly what they mean.

Section 1: What Is Wrong with Distance Alone? (The Phenomenon)	
Explain the matatu puzzle in your own words. Why is it	When Matatu A and Matatu B both leave Kencom Bus Stop and travel exactly 8 km, we might expect them to arrive at the same place. However, this does not happen because distance alone does not tell you which way you are travelling. Distance is a scalar

surprising that both matatus travel 8 km yet end up in completely different places? What does this tell us about the mathematical idea of distance (a scalar quantity)? Define scalar quantity and explain what information it gives — and what information it is missing. **Sentence starter: "When Matatu A and Matatu B both leave Kencom Bus Stop and travel exactly 8 km, we might expect them to arrive at the same place. However, this does not happen because…"**	quantity — it has magnitude (size) only, with no direction attached. For example, saying a matatu travels '8 km' tells you how far it went, but not whether it headed north towards Westlands or south towards South B. Because Nairobi's roads spread out in many directions from the CBD, two travellers can cover identical distances and land in completely different neighbourhoods. This shows us that magnitude alone is not enough to describe a journey completely. We are missing something crucial: direction. The puzzle of the two matatus reveals that we need a new type of mathematical quantity — one that carries both magnitude AND direction.

Section 2: Introducing the Vector — The Complete Description of a Journey	
Define a vector quantity in precise mathematical language. Explain how a vector is different from a scalar. Using the matatu scenario, show how representing each journey as a vector (rather than just a distance) solves the puzzle. In your answer: **– Give the mathematical notation for a vector (e.g., using arrow notation and column vector form).** **– Describe or draw a simple diagram showing the two matatu journeys as vectors on a grid, with Kencom Bus Stop as the origin.** **– Explain what the tail and head of each vector arrow represent in the real-world context.** **Sentence starter: "A vector is a**	A vector is a quantity that has both magnitude and direction. Unlike a scalar, a vector tells us not only how far but also which way. We can write a vector using arrow notation, for example vector AB (with an arrow above), where A is the starting point (the tail) and B is the ending point (the head). In column vector form, we write it as a vertical pair of numbers: the top number shows horizontal displacement (east/west) and the bottom number shows vertical displacement (north/south). For the matatu puzzle, if we place Kencom Bus Stop at the origin O on a grid, then: – Matatu A travels roughly north-west to Westlands. Its journey could be represented as the column vector (−5, 6), meaning 5 km west and 6 km north. Its magnitude is $\sqrt{(5^2 + 6^2)} = \sqrt{(25 + 36)} = \sqrt{61} \approx 7.8$ km ≈ 8 km. ✓ – Matatu B travels roughly south to South B. Its journey could be represented as the column vector (2, −8), meaning 2 km east and 8 km south. Its magnitude is $\sqrt{(2^2 + 8^2)} = \sqrt{(4 + 64)} = \sqrt{68} \approx 8.2$ km ≈ 8 km. ✓ Both vectors have nearly the same magnitude (~8 km), confirming the same distance — yet they point in completely different directions, explaining why the destinations are different. The tail of each vector arrow is at Kencom (the origin), and the head points to the final destination. Direction is what separates the two journeys. Without specifying direction, the description is mathematically incomplete.

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Section 3: Key Properties and Operations of Vectors	
Demonstrate your understanding of vector properties and operations by answering the following: **(a) EQUALITY OF VECTORS: Matatu C travels from Westlands to Pangani, and Matatu D travels from Hurlingham to Kariokor along a parallel road in the same direction and over the same distance. Are these two journeys the same vector? Explain using the definition of equal vectors.** **(b) VECTOR ADDITION: A delivery rider leaves Gikomba Market, rides 3 km east, then turns and rides 4 km north to reach Mutindwa Market. Show how to find the single vector (resultant) that represents the total journey using the triangle law of vector addition. Calculate the magnitude of this resultant vector.** **(c) SCALAR MULTIPLICATION: If the same delivery rider doubles his route, what is the new vector? What changes and what stays the same?** **Sentence starter for (b): "To find the resultant vector of the**	(a) EQUALITY OF VECTORS: Yes, the two journeys represent equal vectors. Two vectors are equal if and only if they have the same magnitude AND the same direction — they do not need to start from the same point. Since Matatu C (Westlands → Pangani) and Matatu D (Hurlingham → Kariokor) travel the same distance in the same direction (let's say north-east), they are equal vectors even though they act on different roads. This is an important idea: a vector is a free vector — it is defined by its magnitude and direction, not its position. (b) VECTOR ADDITION: To find the resultant vector of the two-stage journey, I can use the triangle law of vector addition, which states that if two vectors are placed head to tail, the resultant vector is the one drawn from the tail of the first to the head of the second. Let vector a = (3, 0) [3 km east] and vector b = (0, 4) [4 km north]. Resultant vector r = a + b = (3 + 0, 0 + 4) = (3, 4). Magnitude: $\|r\| = \sqrt{(3^2 + 4^2)} = \sqrt{(9 + 16)} = \sqrt{25} = 5$ km. So the rider's direct displacement from Gikomba to Mutindwa is 5 km. Notice: the total distance ridden is 3 + 4 = 7 km, but the magnitude of the displacement vector is only 5 km. This again shows the difference between scalar distance and vector displacement. (c) SCALAR MULTIPLICATION: If the rider doubles his route, the new vector is 2r = 2(3, 4) = (6, 8). The magnitude becomes $\|2r\| = 2 \times 5 = 10$ km. The direction stays exactly the same (still pointing north-east at the same angle), but the magnitude doubles. Scalar multiplication changes the size of a vector (and can reverse its direction if the scalar is negative) but does not rotate it.

two-stage journey, I can use the triangle law of vector addition, which states that…"	

Section 4: Vectors in Real Life — Beyond the Matatu	
The matatu is just one example. Show that you understand where vectors appear in real Kenyan and global contexts by doing the following: **(a) Identify and explain TWO other real-life situations where vectors are essential (choose from: aviation, weather/wind, sport, agriculture, engineering, or navigation on Lake Victoria).** **(b) For ONE of your chosen contexts, write and solve a short vector problem of your own. Your problem must include a column vector, a magnitude calculation, and a real Kenyan place name or context.** **(c) Give ONE example of a scalar quantity from everyday Kenyan life and explain clearly why it is a scalar and not a vector.** **Sentence starter: "Vectors are not only useful in mathematics classrooms — they appear whenever direction matters in the real world. For example…"**	(a) Vectors are not only useful in mathematics classrooms — they appear whenever direction matters in the real world. For example: 1. AVIATION — Kenya Airways pilots flying from Jomo Kenyatta International Airport (JKIA) in Nairobi to Moi International Airport in Mombasa must account for both the direction they point the aircraft and the wind vector pushing them sideways. The actual path of the plane (its resultant displacement) is the vector sum of the plane's thrust vector and the wind vector. Without vector addition, the plane could drift far off course. 2. FISHING ON LAKE VICTORIA — Fishermen from Kisumu use both the direction of their boat's engine thrust and the current of Lake Victoria when navigating to their fishing grounds. The current exerts a vector force on the boat; adding the boat's velocity vector and the current vector gives the fisherman's true direction of travel. Understanding this helps them reach their nets efficiently and return safely. (b) MY OWN VECTOR PROBLEM — Kisumu Port: A supply boat leaves Kisumu Port and travels with velocity vector v = (4, 3) km/h (4 km/h east, 3 km/h north). A current pushes the boat with vector c = (−1, 2) km/h (1 km/h west, 2 km/h north). Find the resultant velocity and its magnitude. Resultant = v + c = (4 + (−1), 3 + 2) = (3, 5) km/h. Magnitude = $\sqrt{3^2 + 5^2} = \sqrt{9 + 25} = \sqrt{34} \approx 5.83$ km/h. The boat's actual speed is approximately 5.83 km/h in the direction described by the vector (3, 5). (c) SCALAR EXAMPLE — The temperature in Nairobi on a typical March afternoon is 26°C. This is a scalar quantity because it has magnitude only (26 degrees) and no direction is associated with it. You would not say 'the temperature is 26°C heading north.' Compare this with wind speed AND direction (e.g., a 20 km/h north-easterly wind) — that is a vector because direction is essential to its meaning.

Section 5: Answering the Driving Question — Your Full Conclusion

Now bring everything together. Write a full, precise answer to the driving question: **"Why do two matatu journeys of exactly the same distance from Nairobi's CBD end up at completely different destinations — and what mathematical tool do we need to describe a journey completely?"** **Your conclusion must:** **– Directly and fully answer both parts of the driving question.** **– Use correct mathematical vocabulary (scalar, vector, magnitude, direction, displacement, resultant, column vector).** **– Reference at least one calculation or piece of mathematical evidence from your earlier sections.** **– Explain why vectors are a more powerful mathematical tool than scalars for describing journeys (and other real-world phenomena).** **– End with a sentence that connects back to the original map of Nairobi and the two matatus.** **This is your strongest piece of writing in the unit — be precise, be mathematical, and be clear.**	Two matatu journeys of exactly the same distance from Nairobi's CBD end up at completely different destinations because distance is a scalar quantity — it records only magnitude (how far) and contains no information about direction (which way). Since Nairobi's roads extend outward from Kencom Bus Stop in all compass directions, an 8 km journey to the north-west leads to Westlands, while an 8 km journey to the south leads to South B. The scalar '8 km' is identical for both routes, yet the destinations are poles apart. This proves that magnitude alone is mathematically insufficient to describe a journey. The mathematical tool we need is the vector. A vector is a quantity that has both magnitude and direction. In column vector form, the Westlands journey can be written approximately as (−5, 6) — representing 5 km west and 6 km north — with magnitude $\sqrt{5^2 + 6^2} \approx 7.8$ km ≈ 8 km. The South B journey can be written as (2, −8) — 2 km east and 8 km south — with magnitude $\sqrt{2^2 + 8^2} \approx 8.2$ km ≈ 8 km. Both vectors confirm the same approximate distance, but their directions are completely different, which is precisely why the destinations differ. Vectors are a more powerful tool than scalars because they encode the complete information needed to determine an endpoint: not just how far you travel, but in which direction. Operations on vectors — such as addition (combining stages of a journey into a resultant) and scalar multiplication (scaling a journey up or down) — allow us to model complex real-world movement mathematically. Scalars can only tell us how much; vectors tell us how much and where to. Looking back at the Nairobi street map that began this unit, the directional arrows on the roads and the one-way signs were already communicating vector information — direction was always part of the journey. The map did not just show distances; it showed paths with orientation. Mathematics gives us the precise language — the vector — to capture exactly what those road signs were always telling us: that to describe any journey completely, direction is not optional. It is essential.

RUBRIC			
Criterion	**Excellent (4)**	**Proficient (3)**	**Developing (1–2)**

Understanding of Scalar vs. Vector (Conceptual Foundation)	Precisely defines both scalar and vector with correct mathematical language. Clearly explains that scalars have magnitude only and vectors have both magnitude and direction. Connects this distinction directly and explicitly to the matatu phenomenon with specific reference to the 8 km journeys to Westlands and South B.	Correctly defines scalar and vector and explains the key difference. Connects the distinction to the matatu scenario but the explanation may lack full precision or specificity (e.g., does not name both destinations or specify the 8 km detail).	Shows partial understanding — may correctly define one of the two terms but confuses or omits the other. Connection to the matatu phenomenon is weak, vague, or missing. Mathematical vocabulary is limited or used incorrectly.
Vector Notation and Representation	Consistently and correctly uses arrow notation and column vector form. Accurately represents at least two vectors on a described or drawn grid with Kencom as origin. Correctly identifies the tail (starting point) and head (endpoint) of vector arrows. All notation is precise and consistent throughout.	Uses arrow notation and column vector form correctly in most cases. Represents at least one vector on a grid with the origin identified. Minor inconsistencies in notation do not significantly affect meaning. Tail and head of vector arrows are identified.	Attempts to use vector notation but makes repeated errors (e.g., confuses row and column vectors, omits arrow notation, or misidentifies the origin). Grid representation is absent or significantly incorrect. Notation is inconsistent or unclear.
Vector Operations (Addition, Magnitude, Scalar Multiplication)	Correctly applies the triangle law of vector addition to find a resultant vector. Accurately calculates magnitude using the Pythagorean theorem with full working shown. Correctly applies scalar multiplication and explains what changes (magnitude) and what stays the same (direction). All calculations are accurate and clearly linked to a Kenyan real-world context.	Correctly performs vector addition and calculates magnitude with working shown. Applies scalar multiplication correctly. One minor arithmetic or conceptual error may be present but does not undermine overall understanding. Context link is present but may be brief.	Attempts vector addition or magnitude calculation but makes significant errors in method or arithmetic. May confuse distance (scalar) with magnitude of displacement vector. Scalar multiplication is attempted but the effect on direction and/or magnitude is not correctly explained. Working is incomplete or missing.
Application to Real-Life Kenyan Contexts	Identifies two distinct, well-explained real-life contexts where vectors are essential (e.g., aviation at JKIA, fishing on Lake Victoria, Rift Valley farming). Creates an original, correctly solved vector problem using a Kenyan place name, including a column vector and magnitude calculation. Clearly explains one scalar example from everyday Kenyan life with justification.	Identifies two real-life contexts with reasonable explanation. Creates a vector problem that is mostly correct; minor errors in calculation or setup. Scalar example is given with some justification. At least one Kenyan place name or context is used.	Identifies one or two real-life contexts but explanations are vague or do not clearly explain why vectors (not scalars) are needed. Vector problem is absent, incomplete, or contains major errors. Scalar example may be confused with a vector quantity, or justification is missing.
Quality of Final Answer to the Driving Question	Directly and completely answers both parts of the driving question with precision. Correctly uses all required vocabulary (scalar, vector, magnitude, direction, displacement, resultant, column vector). Cites at least one specific	Answers both parts of the driving question adequately. Uses most required vocabulary correctly. References mathematical evidence, though it may not be a specific calculation. Explains the advantage of vectors over scalars.	Attempts to answer the driving question but addresses only one part, or the answer is vague and lacks mathematical precision. Required vocabulary is largely absent or misused. Little or no mathematical evidence is cited. The

	calculation as evidence. Clearly articulates why vectors are more powerful than scalars for describing journeys. Conclusion explicitly reconnects to the Nairobi map and the two matatus, creating a coherent and satisfying close to the unit.	Reconnects to the matatu scenario, though the connection to the original map may be brief or implicit.	connection to the matatu phenomenon and the Nairobi map is missing or superficial.